Lee Kuan Yew School of Public Policy

Final Report

PP5105 Cost Benefit Analysis in Public Policy

A Cost-Benefit Analysis of the Auckland Light Rail

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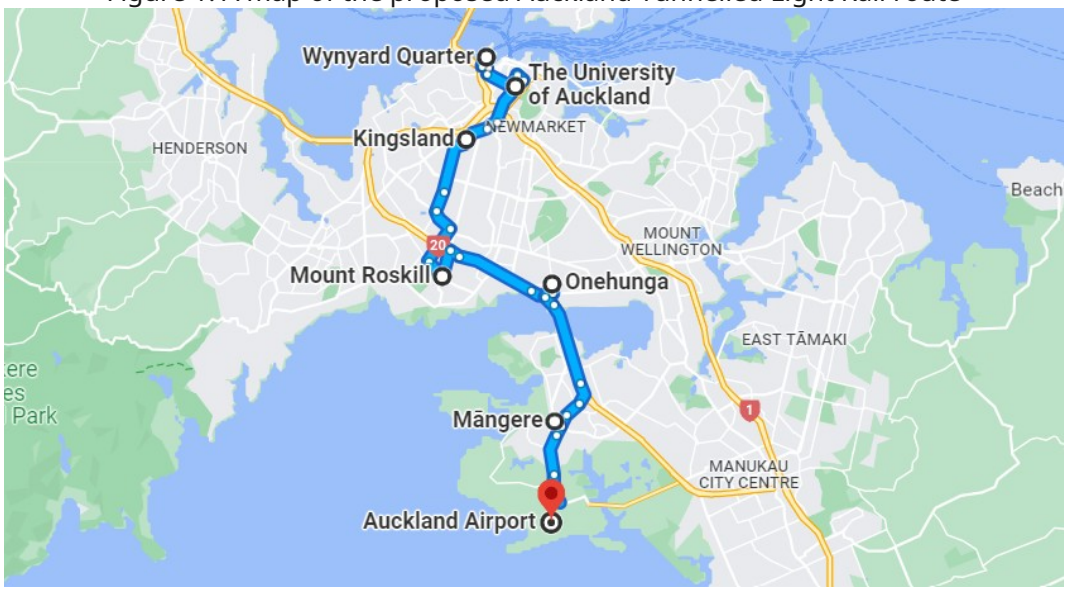
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1. Introduction

Auckland, fondly known to most as the "City of Sails", has earned the dubious distinction of being the "City of Snails" in the last decade due to worsening congestion (Henry, 2019). This has ratcheted up the pressure on the Auckland Council to increase the provision of public transport solutions that could reduce the number of vehicles on the road.

An idea which has gained traction of late is the Auckland Tunnelled Light Rail (ATLR), a proposed 24 km route with both tunnelled and non-tunnelled portions which will connect the Central Business District to South Auckland (see Figure 1). As the largest transport project negotiated by the government through a non-competitive process (Coughlan, 2021) and a key component of Prime Minister Jacinda Ardern's 2017 election manifesto (RNZ, 2017), it has garnered a large amount of public attention. In particular, the Indicative Business Case released by the Auckland Light Rail team was the subject of much criticism, as the Benefit Cost Ratio was only slightly above 1 (Auckland Light Rail, 2022). Moreover, having a BCR which was larger than 1 was largely dependent on the possibility that wider economic benefits[1] would materialise and urban intensification projects would be completed (Welch, 2022).

Figure 1. A map of the proposed Auckland Tunnelled Light Rail route



The flak that the ATLR attracted was unsurprising because there was no transparency regarding how the final NPV value for the ATLR in Figure 2 was estimated. For instance, while time savings to commuters were identified as a benefit from the light rail investment, no workings about how these time savings were quantified and monetised were shown. As such, this study aims to re-estimate the benefits and costs of the ATLR,[2] including those which were not accounted for in the Auckland Light Rail team's report, and evaluate whether the ATLR can indeed pass the cost-benefit test.

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1 An example of wider economic benefits was described as the gains from companies being located in closer proximity to each other. If WEBs were not considered, the BCR would only be 0.5, and the NPV of benefits would have decreased by $5.76 bil.

2 Note that the study compared three different forms of rapid transit - Light Rail, Light Metro and the Tunnelled Light Rail (ATLR). Given the word limit of this report and that ATLR was the chosen option, this report only performs a CBA of the ATLR.

This report will evaluate the ATLR over a 60 year time period from 2021 to 2080, and use a discount rate of 4% for comparability with the Auckland Light Rail team's (2022) report. Even though construction has yet to start in 2022, this study assumes that construction commenced in 2021 while ATLR operations begin in 2031 to facilitate comparisons with their estimates. Lastly, all estimates are reported in 2021 NZD.

Figure 2. PV benefits, PV costs and NPV of the ATLR provided by the Auckland Light Rail team (2022)

	Light Rail	Light Metro	Tunnelled Light Rail
Traditional transport benefits (NPV) (M)	$3,747	$6,063	$5,278
Wider economic benefits (NPV) (M)	$3,989	$6,988	$5,760
Total Benefits (NPV) (M)	$7,736	$13,051	$11,038
Total Cost (NPV) (M)	$7,141	$11,196	$10,362
BCR (Transport benefits only)	0.5	0.5	0.5
BCR (TOTAL benefits)	1.1	1.2	1.1
Project NPV (M) (with WEBS)	$595	$1,855	$676

mation Act

2. Who has standing?

This cost-benefit analysis is conducted at the national level. As the majority of the project's capital expenditure, which amounted to 83% of the reported costs in the Auckland Light Rail team's (2022) report, will be funded by the national government (Smith, 2021). Restricting standing to the Auckland region is not meaningful as it will largely result in a positive net present value. Stakeholders which have standing in this CBA include light rail commuters, domestic producers, the national and local government, and third-parties (such as residents along the route, and other road users). Foreigners, which account for 6.25% of the population (Collins, 2020),[3] are excluded. The global community also has no standing in this CBA, which implies that benefits from any reductions in greenhouse gases will not be counted. The Auckland Light Rail team did not state who had standing and who did not in their report.

3. Identification of benefits & costs

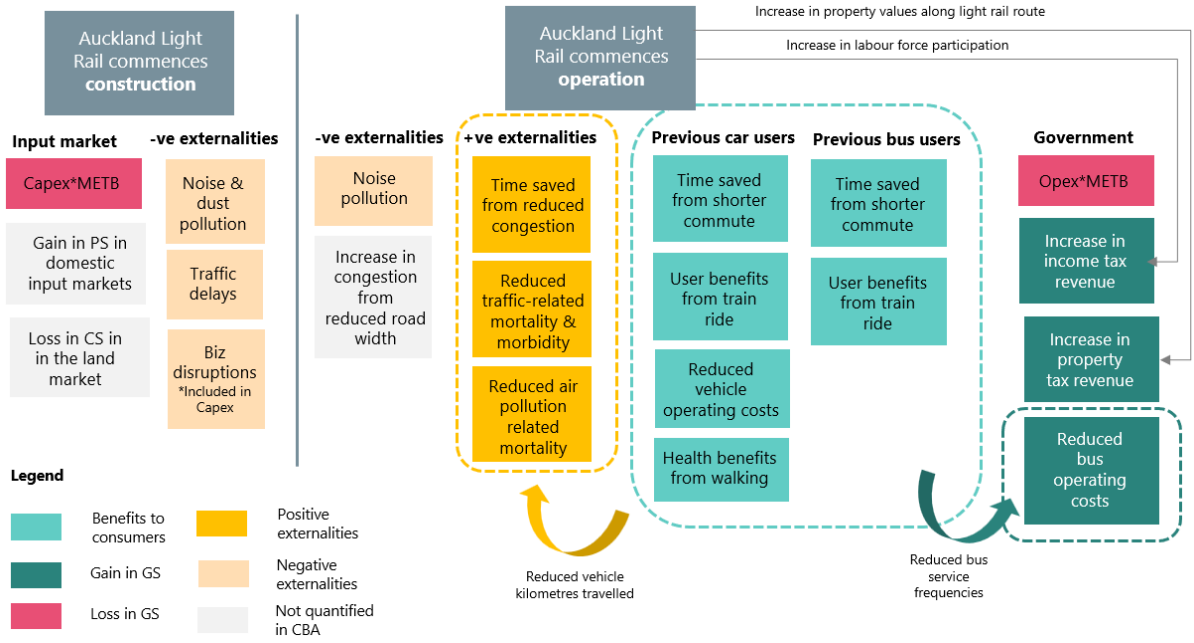
The logic model shown in Figure 3 provides an overview of the benefits and costs from the ALR project. When **construction** of the LR is underway, changes in social surplus occur in the input market and there are negative externalities imposed on third parties. When the light rail commences **operations**, there are surplus changes in the output, input and distorted secondary markets, while both positive and negative externalities are experienced by third parties. Calculations for selective benefits and costs are elaborated on in this section, while calculations for the remaining benefits and costs are shown in the accompanying spreadsheet.

3 6.25% of those living in New Zealand only have a temporary visa. Among the remaining 93.75% of NZ's population, there is no breakdown of the number of citizens, permanent residents and Residence Visa holders provided. Since Residence Visa holders and PRs have largely similar privileges (INZ, n.d.) and also because of data constraints, both groups are given standing in this CBA.

Identification of benefits in the primary output market

Light rail commuters who were previously driving or taking the bus for their commute will benefit from **a reduction in travel time**, which is a well-established outcome from the introduction of public transport interventions (ACIL Tasman, 2011). Academic literature (Grimaldi et al., 2010) and cost-benefit analyses conducted by other governments (Major Projects Canberra, 2019) suggested that commuters gain utility from the experience of riding the train, which is termed as **user benefits**. This is because the train ride is perceived to be more comfortable than car and bus rides, due to reduced jerkiness and greater freedom of movement during the transport journey. Additionally, for light rail commuters who were previously commuting via car, the switch to the light rail **reduces their vehicle operating costs.** As two-thirds of passengers in Auckland walk to the train station (Auckland Transport, n.d.), this group of commuters will also witness some **improvements in health** from an increase in physical activity. These health benefits are not counted for light rail commuters who were previously taking the bus, as it is assumed that they were already walking to the bus stop.

Figure 3: A logic model of the benefits and costs from the operation of the ALR



Identification of benefits in the input market

As the government will procure a substantial amount of construction services from the domestic market for the ATLR project (ALR, 2022), there will be **some gains in producer surplus (PS)** in this input market. The actual decrease in social surplus in the input market could be lower than the reported fiscal outlay of the project (refer to Annex B). However, this report does not quantify and monetise the gain in PS in the input markets due to data constraints.

Identification of benefits in the secondary markets

There would be substantial positive externalities to third-parties from the ATLR project. Firstly, one would expect **time savings to other road users** from reduced congestion. Fageda's (2021) study of light rail projects in 98 European cities across 13 countries from 2008 to 2019

discovered a 7% reduction in congestion. Secondly, the same study found that there was a 3% reduction in air pollution when light rail projects were introduced, which is associated with **a fall in premature deaths** (Huang et al., 2018). Thirdly, a reduction in vehicle kilometres travelled results in **fewer road accidents** (Yasin et al., 2021), and the consequent reduction in mortality and morbidity should also be valued. Fourthly, Lee's (2017) study of the opening of two light rail lines in Los Angeles showed that after accounting for the first and last mile bus journeys to the train station, one would still expect to observe a net decrease in ridership for bus trips. This would translate into a scaling down of bus services, and hence **a reduction in bus operating costs** incurred by the transport authorities.

As the secondary markets for labour and property are taxed, some surplus changes in these markets are not captured in the primary market for light rail services (Boardman et al., 2018). Hence, **increases in income tax and property tax revenues** from an expected boost to labour force participation (Pons et al., 2015) and rise in property values (Hess & Almeida, 2007) with the introduction of a light rail should be counted as gains in government surplus.

Comparison of benefits included in both reports

Table 1 below compares the benefits that were included in the ALR (2022) report and this study. This report included most of the benefits outlined in the ALR report, except the gain in utility for ATLR users and reliability benefits due to data constraints. Including both agglomeration benefits and time savings would appear to be double counting, so the former was excluded. As the ALR team's (2022) report offered no clarity on how the implementation of the ATLR would reduce the marginal costs of production, this benefit was not included.

Benefits in both reports	Benefits in this report, but not ALR's report	Benefits in ALR's report, and reason for exclusion from this report
User experience benefits	Time savings	Gain in utility from ATLR users: Could not be estimated in this report as choke price is not known
Reduced vehicle operating costs	Reduced bus operating costs	Reliability benefits: Not included in this report as it is difficult to quantify
Time savings to road users from decongestion	Fall in premature deaths due to reduced air pollution	Agglomeration benefits: Productivity gains from being located in closer proximity: Not included due to double counting
Improvement in health of commuters	Gain in producer surplus in the input market	Increase in production due to reduction in marginal costs from light rail infrastructure: Benefit is not clearly defined
Reduced mortality and morbidity from reduced accidents	Increase in property tax revenues	
Increase in income tax revenue		

Identification of costs in the primary input market

During **construction** of the ATLR, **capital expenditure** (Capex) will be incurred. As this project has a social procurement objective (ALR, 2022), it will hire from a pool of unemployed workers which have a true opportunity cost that is lower than the wages they are paid (Boardman et al.,

2018). Since a breakdown of the capital expenditure estimates were not provided, it is not possible to provide a scaled down amount of the true opportunity cost of labour. The estimate of capital expenditure should be multiplied by one plus the **marginal excess tax burden** as there are efficiency losses in the labour market when income tax is raised to fund this project (ibid). This is especially since the New Zealand Government raises the largest share of its tax revenue from personal income taxes (Inland Revenue, 2017). As the supply of land is relatively inelastic in Auckland (Taylor, 2010), there will be some **loss in consumer surplus** when the government acquires land for the construction of the ATLR (refer to Annex B). This introduces the possibility of underestimating the loss in social surplus in the input market, which is not feasible to estimate here due to data constraints. Upon the commencement of the ATLR **operations** in 2031, **operational expenditures** (Opex) will be incurred annually, which should also be multiplied by the **marginal excess tax burden.**

Identification of costs due to negative externalities

The **construction** of railway lines will result in **noise** (Marshall Day Acoustics, 2020) and **dust pollution** (Wang et al., 2022) to households living along the route, with corresponding health effects that should be accounted for. Other adverse impacts include **increased travel time** due to reduced road access and **disruptions to businesses**. All the abovementioned impacts are negative externalities which result in welfare losses, and should be monetised, although business disruptions were already accounted for in the ALR team's Capex estimates.

After the ATLR is **operational** in 2031, households living along the non-tunnelled portion of the ATLR will continue to experience **noise pollution** (MDA, 2020). Any **increase in congestion from reduced road width** is expected to be minimal. This is because the non-tunnelled portion of the ATLR runs separately to State Highway 20, which is the main thoroughfare from central to South Auckland (Kohere, 2022).

Comparison of costs included in both reports

Table 2 below compares the costs that were included in ALR's (2022) report and this study. Capital and operating expenditures were included in both reports. It was assumed that the Marginal Excess Tax Burden was not a component of the Capex and Opex estimates, as there was no mention of this in ALR's report. Losses in consumer surplus due to the government's acquisition of land for the ATLR also appeared to be neglected. While "noise, dust, vibration, access restrictions" were recognised as adverse community impacts in ALR's (2022, p. 106) report, it was unclear whether they were a component of the cost estimates.

Costs in both reports	Costs in this report, but not ALR's report
Capital expenditure (Capex) (including business disruptions), Operational expenditure (Opex)	Loss in consumer surplus when land is acquired Marginal Excess Tax Burden associated with Capex and Opex Negative externalities during construction and operation, such as noise/ dust pollution, delays in travel time and business disruptions

4. Quantifying benefits and costs

While there was clarity about *what* the Auckland Light Rail team chose to quantify and monetise, no information was provided about *how* these identified benefits and costs were monetised. All

estimates in this section, except for Capex and Opex, were thus informed by the authors' own research, and served to double check the ALR's report NPV value.

Constructing the base case

It was assumed that Auckland's population will increase by 1.8% annually (Statistics NZ, 2021a), while 51.6% and 10.8% of the population will commute by driving and taking the bus respectively (ALR, 2022). This means that there would have been between 0.7 - 2 mil drivers and 0.18 - 0.42 mil bus commuters from 2021 to 2080 in the base case. Congestion levels over the next 60 years were predicted based on historical trends observed from the TomTom Index for Auckland (n.d.).This resulted in an increase in travel times by car and bus from 51 to 58.5 mins, and 71 to 81 mins respectively between 2031 to 2080 in the base case.[4]

Estimating ATLR patronage

Estimating ATLR patronage is fundamental to the derivation of benefits for this CBA. It was assumed that ATLR patronage would be derived from two sources: Commuters who would have driven and taken the bus in the base case. Minimal substitution between other train services and the light rail is expected to occur (ibid), while demand for ATLR trips from other transport modes (such as cycling) will be insignificant.[5] Based on Mulalic and Rouwendal's (2020) simulation, it is assumed that 2.5% of those who would have been driving in the base case will shift to the light rail. Their paper was selected to inform this study's estimates, as the Copenhagen Metropolitan Area shares many similar characteristics with Auckland.[6] According to a meta-analysis by Fearnley et al. (2018), the cross modal elasticity between the bus and light rail is 0.15. Based on calculations shown in Annex C, one would expect 6.86% of those who were taking the bus in the base case to use the light rail. With these assumptions, it is estimated that there will be 24.1 mil ATLR trips in 2031 which will ramp up to 57.8 mil in 2080.

Quantifying benefits in the primary output market

Table 3 below provides a summary of the calculations performed to quantify the respective benefits, which is followed by a more detailed discussion of Benefits #1 and #3.

Who?	Item	Methodology
For all light rail commuters	1. Time saved for light rail commuters	Change in travel time per trip relative to base case (hrs) x Weighted hourly opportunity cost of time x Annual number of ATLR trips
	2. User benefits (See Annex D for	Willingness to pay for a better user experience on the train of NZD 1.95 x Annual number of ATLR trips

4 Specifically, travel times via car for the current year was estimated using Google Maps, and journey times via bus for the current year was obtained from the Auckland Transport App. Journey times from 2031 to 2080 were then predicted based on the ratio of the current congestion level to the predicted congestion level.

5 Those who currently commute by walking or jogging (9.2%) are unlikely to switch to the light rail, as it is presumed that their commuting distance is minimal. Other transport modes, such as taking the ferry (0.6%) or cycling (1.2%) will not contribute substantially to ATLR patronage relative to existing car and bus commuters.

6 There are numerous similarities between Copenhagen and Auckland. Both contribute approximately 40% to the national GDP, have geographical areas of between 1,000-2,000 square kilometres and have a population of 1.5-1.6 mil (ATEED, 2019; Mulalic & Rouwendal, 2020; Statistics NZ, 2021b; World Population Review, n.d.).

	benefit transfer calculations)	
For light rail commuters who previously drove	3. Health benefits	Average distance walked to train station x Gain in utility from health benefits per km x Proportion of previous drivers who walk to train station x Annual number of ATLR trips
	4. Reduced vehicle operating costs	Number of drivers who switched to ATLR x Annual vehicle operating costs of NZD 2053.53 obtained from the New Zealand Transport Agency (2019)

Time savings were obtained by subtracting the new commuting time from Central Auckland to South Auckland from the commuting times mentioned in the base case. The new commuting time (50.6 mins) is the sum of the in-vehicle journey time using the light rail (32 mins) (ARL, 2022), the average waiting time (2.5 mins) and the average walking time (16 mins).[7] This resulted in time savings of 0.5 to 7.9 mins and 18.4 to 30.4 mins for previous car and bus commuters respectively from 2031 to 2080. This was then multiplied by the hourly opportunity cost of time, which was a weighted average of the median wage, unemployment benefit rate and pension rate (see Annex E), and the number of ATLR trips.

Health benefits were monetised by first computing the average distance walked to the train station, which was 1.1 km (Auckland Council, 2013). The gain in utility from walking per kilometre was 4.64 in 2021 NZD according to the New Zealand Transport Agency's (2021) manual. Two-thirds of train commuters in Auckland walk to the train station (Auckland Transport, n.d.). These three parameters were multiplied by the number of trips by ATLR passengers who switched from driving to obtain the monetised health benefits per year.

Quantifying benefits in the secondary market

Table 4 summarises the calculations performed to quantify the respective benefits, which is followed by a discussion of how the expected reduction in vehicle kilometres travelled and the change in travel time for those who continue to use the road were estimated (Benefit #5).

Who?	Item	Methodology
Other road users	5. Time saved for other road users	Change in travel time per trip relative to base case (hrs) x Weighted hourly opportunity cost of time x Annual number of car/ bus trips in Central and South Auckland
Society	6. Reduced mortality from road accidents	Ratio of mortality from road accidents to Vehicle Kilometres Travelled from Ministry of Transport (n.d.) x Reduction in vehicle kilometres travelled x Value of a statistical life (4.52 mil; MOT, 2019)
	7. Reduced morbidity from road accidents	Ratio of injuries from road accidents to Vehicle Kilometres Travelled from Ministry of Transport (n.d.) x Reduction in

7 The average waiting time is half of the ATLR frequency of 5 minutes (ALR, 2022), and the average walking time was derived from the average walking distance to the train station multiplied by the average walking speed.

		vehicle kilometres travelled x Social cost of injuries (25k for minor and 475k for serious injuries; MOT, 2019)
	8. Reduced mortality from air pollution	Ratio of air pollution related to mortality to Vehicle Kilometres Travelled (Briggs et al., 2016) x Reduction in vehicle kilometres travelled x Value of a statistical life
Government	9. Reduced bus operating costs	Reduction in bus services in Kilometres[8] x Bus operating costs per km[9]
	10. Increase in income tax revenue	Increase in labour force participation of 2% for individuals living along the ATRL[10] x Average median wage for the two lowest income quintiles[11] x Relevant tax rate of 17.5%
	11. Increase in property tax revenue	Average change in property tax value in Auckland[12] x Property tax rate of 0.22%[13] x Number of properties along ATLR route (obtained from the ALR report)

As there is a mode shift from cars and buses to the ATLR, there will be a **reduction in Vehicle Kilometres Travelled**. The decrease in VKTs attributed to commuters who would otherwise be driving was obtained by assuming that each car travelled 10,000 km per year (MOT, 2015), and multiplying this figure by the number of drivers who shifted to the ATLR. The decrease in VKTs attributed to a switch from bus to ATLR was obtained by computing the reduction in bus services and multiplying this figure by the length of a bus service, which was assumed to be 28 km (length of the ATLR corridor). This amounted to a total decrease of VKTs ranging from 436 million in 2031 to 1 billion in 2080, which then formed the basis for deducing the reduced number of vehicle-related deaths (i.e. Benefits 6, 7, and 8) as well as reduced bus operating costs (i.e. Benefit 9).

The fewer number of vehicles on the road also implied that there would be reduced congestion, which would lead to **time savings** for those who continue to use the road. To deduce the new congestion level, first, a scatterplot of historical congestion levels on vehicle kilometres travelled was plotted and a trend line was fitted (see Annex F). Secondly, the reduction of VKTs attributed to the ATLR was subtracted from the number of VKTs in the base case. Thirdly, the new VKT value was substituted into the linear equation obtained from the scatterplot to acquire the expected congestion levels from 2031 to 2080 if the ATLR was introduced. Fourth, the new

8 The proportion of bus services to bus commuters (Lambert, 2019), 0.0867 as of 2019 –pre-pandemic–, was assumed to be constant throughout the evaluation period. Then, the expected number of bus services in the base case was compared to that with the ATLR.

9 The operating cost per km was taken from Beca, 2012.

10 Using a difference-in-difference approach, Pons et al's (2015) study found that there was an increase in employment rate of 2% for those residing along the Metro station in Copenhagen when it became operational.

11 It is assumed that an increase in labour force participation will be largely attributed to lower income workers, as middle and high income workers were unlikely to be deterred from seeking employment due to a lack of public transport options (MRCagney, 2020). As such, the average of the median wage for the two lowest income quintiles, and not the median wage, was used to derive the amount of additional tax revenue collected.

12 It was assumed that property values would increase by 2.5% relative to the base case, based on studies of movements in property prices in the US upon the extension of metro lines (Hess & Almeida, 2007).

13 Based on property rates as per Auckland Council (n.d.b.)

travel time for those who continue to drive and take the bus was computed using this set of congestion levels. Lastly, the new travel time was compared to the previous travel times in the base case to arrive at the finding that there would be an average time saving of 1.04 mins and 1.44 mins per car and bus trip when the ATLR became operational. These decongestion benefits were assumed to only accrue to commuters in Central and South Auckland, which represented 50% of Auckland's population (see Annex G).

Quantifying costs

Estimates of **capital and operational expenditures** of 8.6 bil and 1.7 bil in Net Present Value terms and 2021 NZD were obtained from the ALR's (2022) report. Both Capex and Opex values were multiplied by a 0.2 to obtain the marginal excess tax burden (METB), as the recommended METB estimates from the NZ Government is 1.2 (Treasury NZ, 2015).

Additionally, negative externalities were monetised using the methodologies specified in Table 5 below. Firstly, the **delay in travel time during ATLR construction** was obtained by finding the average additional travel time taken if an alternative route had to be used, which amounted to 2 minutes based on Google Maps. Similar to the reasoning in the above section, it was assumed that only 50% of Auckland's commuters will be affected by this delay.

Who?	Item	Methodology
Households along ATLR	Noise pollution during construction (See Annex D for benefit transfer calculations)	Number of households living along the route from ARL's (2022) report x Willingness to pay to reduce noise pollution of NZD 125 per person per year
	Noise pollution during operation	Number of households living along the non-tunnelled portion of the route from ARL's (2022) report x WTP to reduce noise pollution of NZD 125 per person per year
	Dust pollution during construction	Number of households living along the route from ARLs x Average household size x Willingness to pay per person for reduction in dust air pollution based on Akhtar et al.'s (2017) estimates
Other road users	Delay in travel time during construction	Increase in travel time per trip (2 mins) x Weighted hourly opportunity cost of time x Annual number of car/ bus trips for 50% of Auckland

5. Results

As shown in the table below, the Present Value (PV) of all monetised benefits amounted to 19.4 billion in 2021 NZD, whereas the PV of all monetised costs amounted to 17.6 billion in 2021 NZD. This resulted in a Net Present Value of 1.79 billion in 2021 NZD and a Benefit Cost Ratio of 1.10. The two largest benefits were time savings for other road users and time savings for ATLR commuters, corresponding to 44% and 29.3% of total benefits respectively (refer to Annex H for a graphical depiction). The largest components of the cost estimates were capital expenditure and welfare losses due to delays in travel time during the construction of the ATLR, amounting to 58.9% and 15.8% of PV costs respectively. As compared with the ALR team's estimates, the PV of benefits estimated by this report was 8.33 billion higher. This could be because positive

externalities, which accounted for the largest share of benefits in this CBA, were not adequately accounted for in the ALR team's study. Additionally, this analysis also considered two gains in government surpluses, such as the reduction in bus operating costs and the increase in property tax revenues, which were not mentioned in the ALR team's report. Similarly, the PV of costs was 7.22 billion larger in this study, as negative externalities were monetised in this report but did not appear to be included in the ALR team's values.

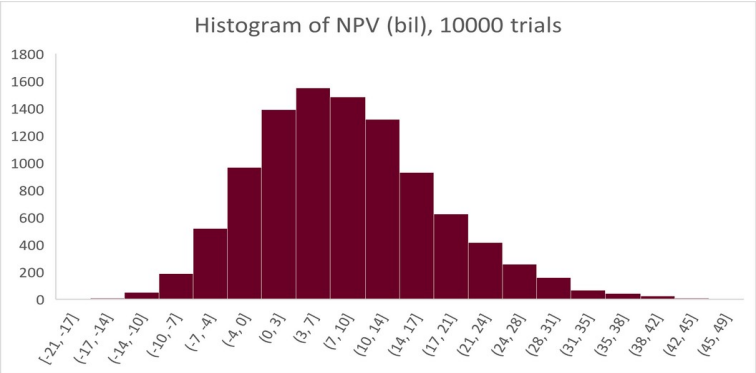
Cost		Component	Total (2021 NZD)
Noise Pollution			451,180,000
1	Noise pollution during construction	180,740,000	
2	Noise pollution during operation	270,450,000	
Dust Externalities			109,290,000
Delay During Construction			2,777,000,000
Capex			10,362,000,000
Opex			1,811,000,000
METB			2,072,000,000
PV Total Costs			**17,583,370,000**
Benefits			
Time Savings for ATLR commuters			5,672,000,000
1	Time Savings for ex-drivers	1,178,800,000	
2	Time Savings for ex-bus commuters	4,493,050,000	
User Experience Benefits for ATLR commuters			806,000,000
Reduced Private Vehicle Operating Costs			840,500,000
Reduced Bus Operating Costs			173,430,000
Health Benefits			993,000,000
Reduced Vehicle-Related Deaths			511,100,000
1	Reduction in Traffic Mortality	200,000,000	
2	Reduction in Minor Injuries	42,670,000	
3	Reduction in Serious Injuries	89,120,000	
4	Reduction in Air Pollution Deaths	179,560,000	
Time Savings For Other Road Users			8,531,100,000

1	Time Savings For Car Users	6,653,150,000	
2	Time Savings For Bus Users	1,877,973,000	
Increase in Income Tax Revenue			596,000,000
Increase in Property Tax Revenue			1,255,000,000
PV Total Benefits			**19,377,000,000**
NPV (Benefits - Costs)			**1,794,000,000**
Benefit Cost Ratio			**1.10**

6. Sensitivity analysis

A Monte Carlo simulation with 10,000 trials was performed using the most important and uncertain parameters, such as the proportion of drivers who switched to the light rail, the value of the statistical life, Capex and Opex, as well as the discount rate. Firstly, the percentage of drivers who would decide to change their mode of transport to the light rail was varied from 2 to 3 percent based on the range provided in Mulalic and Rouwendal's (2020) study. Secondly, the Value of Statistical Life was allowed to vary from 3.93 to 4.52 million, by taking reference to the VSL estimates from other jurisdictions (Australian Govt, 2020; Dolan & Jenkins, 2020). Thirdly, the upper ranges of Capex and Opex estimates were 30% higher than the value in the base case based on historical cost overruns of infrastructure projects (Stubbs, 2022). Lastly, the discount rate was given a lower and upper bound of 4 to 5%, as the Treasury's (n.d.). Guidance suggested that 5% was the appropriate discount rate for infrastructure projects, but the ALR team utilised a discount rate of 4% in their study. A histogram of the results from the Monte Carlo simulation (see Figure 4) revealed that the modal value was approximately 3-7 billion, which was close to the range of this study's NPV of 1.92 billion.

Figure 4: A histogram of a Monte Carlo simulation of this CBA



7. Policy implications

With a Benefit Cost Ratio of 1.10, this report's conclusion is not different from what the ATLR's initial cost benefit analysis discovered. Moreover, the positive modal value from the Monte Carlo simulation provides greater confidence that the Auckland Tunnelled Light Rail project should be given the green light. The scale of benefits and costs in this project was significantly larger than that of the initial analysis, suggesting that the positive and negative impacts of the project on New Zealand's finances and Auckland's population could be larger than initially

anticipated when externalities were accounted for. However, one could not rule out that this could be because of imprecision in the values of parameters underpinning this study. While robust empirical evidence was drawn on to inform the estimates in this analysis, the authors were still subjected to data constraints, such as not being privy to more granular data on current public transport patronage along the proposed light rail route.

8. Conclusion

This report started out with a sceptical stance towards the cost benefit analysis that the ALR team carried out to evaluate the feasibility of the ATLR project to connect Central to South Auckland. The main suspicions with regards to the initial evaluation concerned the failure to include important elements in the analysis, such as positive and negative externalities, and an overall lack of transparency in the estimations. These suspicions appeared to be warranted, as the results indicated that the scale of the benefits and the costs were larger when externalities and other changes in government surplus were accounted for. While the recommendation that the ATLR should be constructed remained the same, this report included a much wider, more detailed and more transparent analysis of the benefits and costs in support of the ATLR. Even though many assumptions were made along the way due to lack of more granular data, this report grounded them in transparent reasonings and recent empirical evidence. This approach is what makes cost benefit analysis an effective decision making tool when evaluating among policy alternatives. Anything less would only be speculation.

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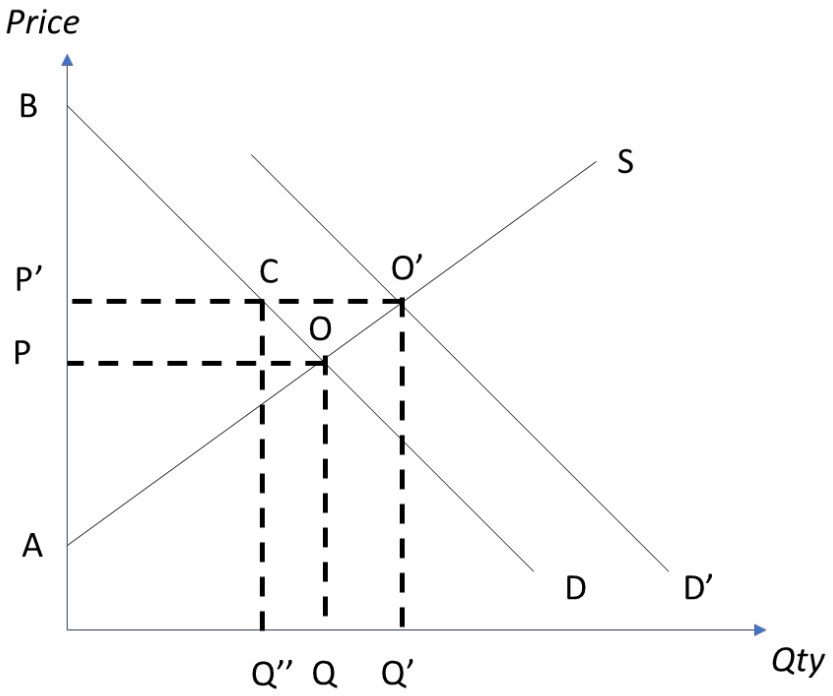
Annex A: Reasons for revision of assumptions between presentation of findings and writing of this report

Item	Change in estimation methodology
No. of drivers	● The team previously assumed that: Number of drivers = Number of households x Individuals per households x Vehicle per capita ● However, as not everyone who owns a vehicle will drive for their daily commute, it is more accurate to use the estimate that 51.6% of Aucklanders drive to work (ALR, 2022).
% of previous bus commuters which switch from bus to light rail	● It was previously assumed that 3.5% of commuters who would have taken the bus in the base case will instead commute via the ATLR. This was based on one study in the United States. ● The authors then found Fearnley et al's (2018) study, which serves as a more reliable source to estimate the mode shift from bus to ATLR as it is a meta-analysis of the cross modal elasticity of different transport modes.
Estimation of bus operating costs	● Based on feedback received during the presentation, now it is assumed that the mode shift from bus to light rail will not reduced the number of buses, but the number of *bus services* provided. ● Based on feedback received from peers during the presentation, the team factored in the demand for bus services from passengers who will make their first and last mile journeys to the train station by bus. ● This was assumed to be ⅓ of light rail passengers, as ⅔ of train commuters in Auckland walk to the train station (Auckland Transport, n.d.) and the number of parking spaces available is substantially outstrpped by demand (Tan, 2018). ● Taking this into account will revise the benefits from reduced bus operating costs downwards.
Negative externalities from dust pollution	● This was previously not considered as a negative externality. Welfare losses from dust pollution are now monetised in the CBA.
Time savings for other road users	● It was previously assumed that all remaining road users will benefit from decongestion. Now, it is only assumed that those living in Central and South Auckland are likely to benefit from decongestion (see Annex G).
Increase in time due to light rail	● It was previously assumed that only those living along CC2M will experience a delay in travel time when the light rail was

construction	being constructed. ● To be consistent with how time savings for other road users were estimated, the delay is now assumed to be experienced by all living in Central and South Auckland.

Annex B: Graphical depictions of changes in social surpluses

Figure 5. Graphical depiction of potential gains in producer surplus in the input markets

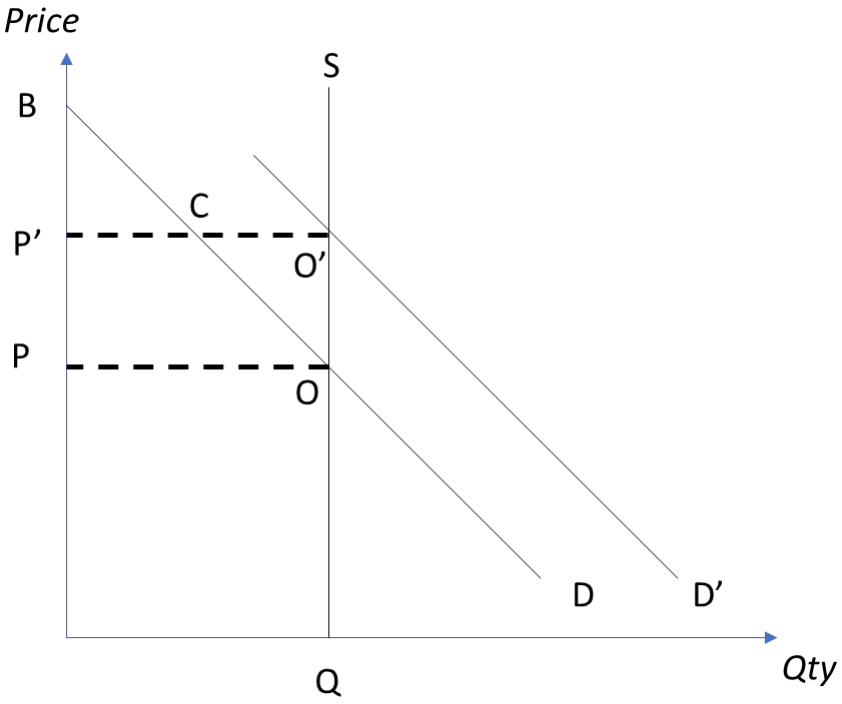


The figure above depicts the change in social surplus in the input market for construction due to demand increase induced by the procurement of supplies by the government for ATLR construction. There is a net gain in social surplus represented by COO', which is not accounted for by changes in social surplus in the primary market. This is described in as follows:

	Before	After	Change in surplus
Change in Consumer Surplus	BPO	BP'C	-P'COP
Change in Producer Surplus	APO	AP'O'	OO'P'P
Net change in social surplus			COO'

The fiscal outlay of the government on this project is CO'Q'Q". However, the true opportunity cost would be CO'Q'Q" minus the change in surplus COO' i.e. COO'Q'Q"

Figure 6. Graphical depiction of potential losses in consumer surplus in the input market for land



The figure above shows that demand for land is inelastic and an increase in demand for land due to procurement of land by the government for construction of the ATLR increases price leading to a decrease in consumer surplus represented by P'COP This is described below, and this decrease in CS is not accounted for by changes in social surplus in the primary market.

	Before	After	Change in surplus
Change in Consumer Surplus	BPO	BP'C	-P'COP
Net change in social surplus			-P'COP

Annex C: Estimating mode shift from bus to ATLR

Item	Estimate	Reference
In-vehicle journey time for bus commutes in 2031	60.6 mins	Auckland Transport app
In-vehicle journey time for the light rail commute	32 mins	Auckland Light Rail (2022) report
Change in in-vehicle journey time by making switch from bus to light rail	-0.457	
Elasticity of bus demand with respect to in vehicle journey time for light rail	0.15	Fearnley et al. (2018)
Reduction of bus demand	-6.86%	

Annex D: Benefit transfer calculations

Benefit transfer of WTP for user experience benefits from Chang and Jung's (2017) study on utility gains from train service improvements

Benefit transfer across time

	Value	Reference
Y base (WTP from Chang and Jung's contingent valuation survey of commuters in 2014)	0.91 USD	Chang and Jung (2017)
Income target (South Korea's GNI per capita in 2019)	29,580 USD	World Bank
Income base (South Korea's GNI per capita in 2014)	24,601 USD	World Bank
Elasticity	0.8	Lecture Slides
Y target (WTP of South Korean commuters at present)	1.06 USD	

*GNI per capita from the year 2019 was used to inform calculations of commuters' WTP in 2021, as national incomes in 2020 - 2021 were affected by the COVID-19 pandemic.
*All values were inflation-adjusted.

Benefit transfer across setting

	Value	Reference
Y base (WTP of South Korean commuters at present)	1.06 USD	Benefit transfer across time
Income target (New Zealand's GNI per capita in 2019)	37,452 USD	World Bank
Income base (South Korea's GNI per capita in 2019)	29,580 USD	World Bank
Elasticity	0.8	Lecture Slides

Y target (WTP of New Zealand commuters at present)	1.68 NZD	After currency conversion and adjusting for inflation from values in 2014 USD

*GNI per capita from the year 2019 was used to inform calculations of commuters' WTP in 2021, as national incomes in 2020 - 2021 were affected by the COVID-19 pandemic.
*All values were inflation-adjusted.

Benefit transfer of WTP for willingness to pay to avoid noise pollution from Istamto et al's (2014) study of 10,464 respondents across 5 European countries

Benefit transfer across time

	Value	Reference
Y base (WTP from Istamto et al's contingent valuation survey of European residents in 2010)	66.5 USD	Istamto et al.. 2014
Income target (EU's GNI per capita in 2019)	27,936 USD	World Bank
Income base (EU's GNI per capita in 2010)	28,944 USD	World Bank
Elasticity	0.8	Lecture Slides
Y target (WTP of European residents at present)	64.64 USD	

*GNI per capita from the year 2019 was used to inform calculations of commuters' WTP in 2021, as national incomes in 2020 - 2021 were affected by the COVID-19 pandemic.
*All values were inflation-adjusted.

Benefit transfer across setting

	Value	Reference
Y base (WTP of European residents at present)	64.64 USD	Benefit transfer across time
Income target (New Zealand's GNI per capita in 2019)	34,497 USD	World Bank

Income base (EU's GNI per capita in 2019)	27,936 USD	World Bank
Elasticity	0.8	Lecture Slides
Y target (WTP of New Zealand commuters at present)	125 NZD	After currency conversion and adjusting for inflation from values in 2010 USD

*GNI per capita from the year 2019 was used to inform calculations of commuters' WTP in 2021, as national incomes in 2020 - 2021 were affected by the COVID-19 pandemic.
*All values were inflation-adjusted.

Annex E: Weighted hourly opportunity cost of time

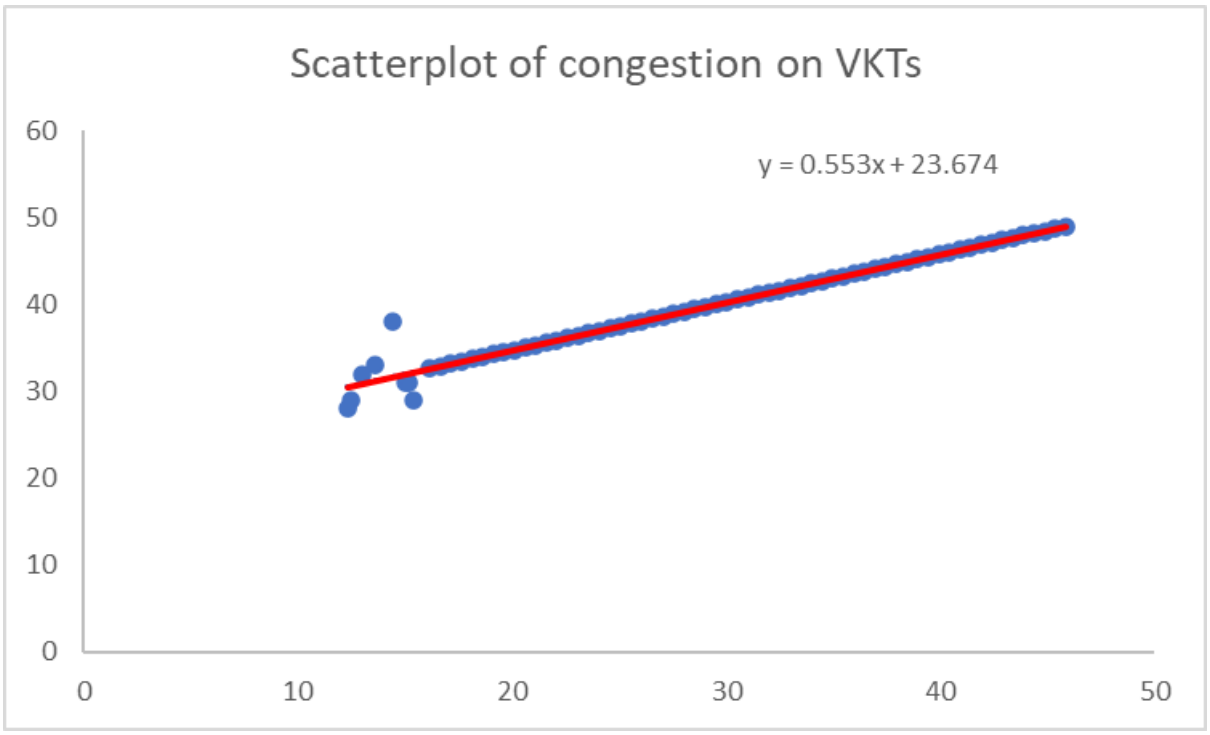
The weighted hourly opportunity cost of time was derived from the following equation, so as to account for heterogeneity in the opportunity cost of time across different population sub-groups:

Weighted hourly opportunity cost of time = (Proportion in employment x Median hourly wage) + (Proportion who are unemployed x Average hourly unemployment benefit rate) + (Proportion who are not in the labour force x Average hourly pension rate)

Item	Estimate	Reference
Proportion in employment	67.8%	Auckland Council (2019)
Proportion in unemployment	4.2%	Auckland Council (2019)
Proportion who are not in the labour force	28%	Auckland Council (2019)
Median hourly wage in Auckland (2021 NZD)	46.71	Statistics NZ (2018)
Average annual increase in real median hourly wage	1.9%	ECA (2021)
Average hourly unemployment benefit rate	6.27	WINZ (n.d.a.)
Average hourly pension rate	12.61	WINZ (n.d.b.)

Annex F: Estimation of relationship between congestion and vehicle kilometres travelled

Figure 7. A scatterplot of congestion levels on vehicle kilometres travelled, with a linear trendline fitted

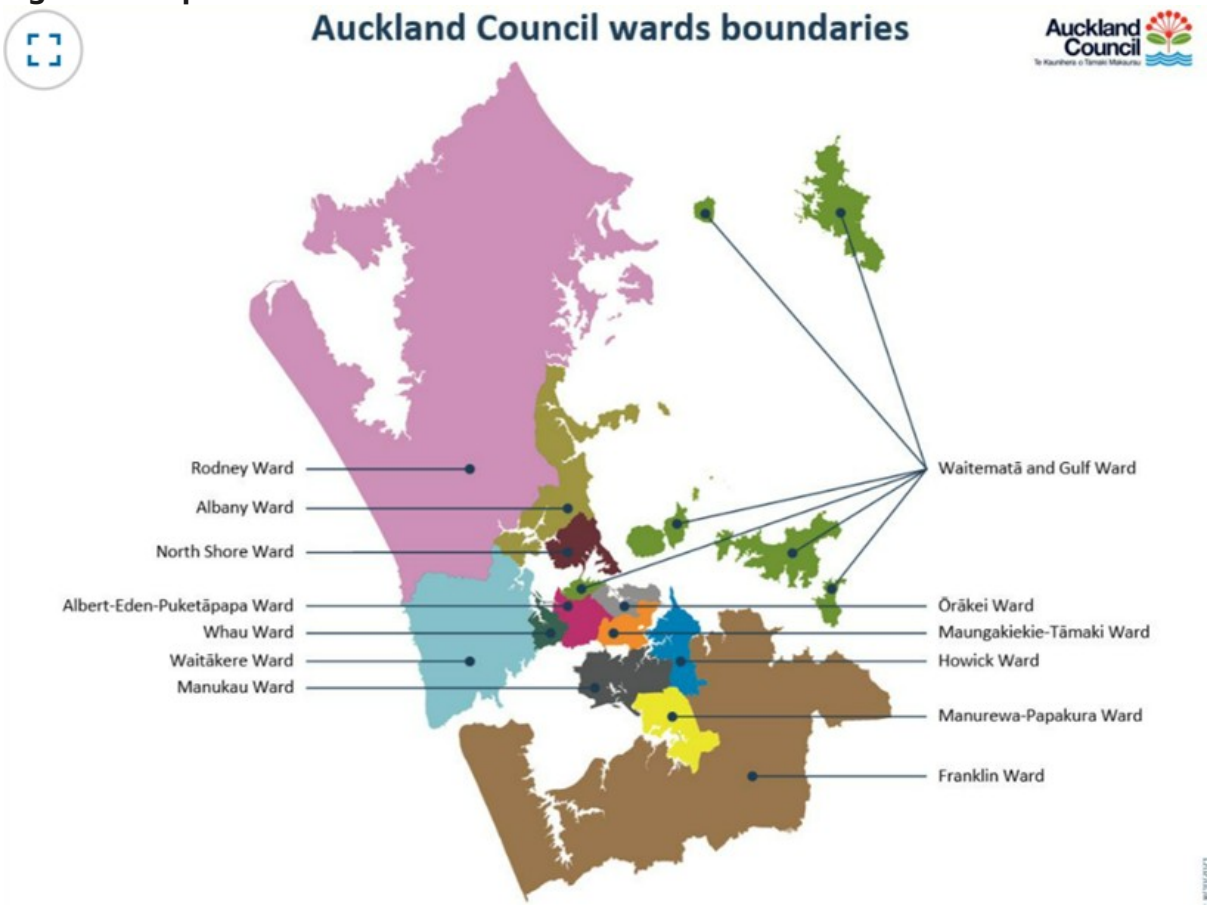


Annex G: Identifying the "catchment area" where decongestion benefits will be accrued

It was assumed that only individuals living in Central and South Auckland will be affected by delays in travel time during the construction of the light rail, and benefit from decongestion when the ATLR becomes operational. Individuals living in North, West and the fringes of South Auckland are unlikely to benefit from ATLR.

By identifying the Auckland Council wards which are located within Central and South Auckland (Figure 9) and obtaining the number of residents living in these wards (Figure 8), the team deduced that approximately 50% of Auckland's residents live in Central and South Auckland. However, this would be an underestimation of the actual number of Auckland residents who are affected by congestion and decongestion due to the ATLR, as some individuals living in North and West Auckland would nevertheless commute to Central and South Auckland for work.

Figure 8. Map of Auckland



Reference: Auckland Council (n.d.)

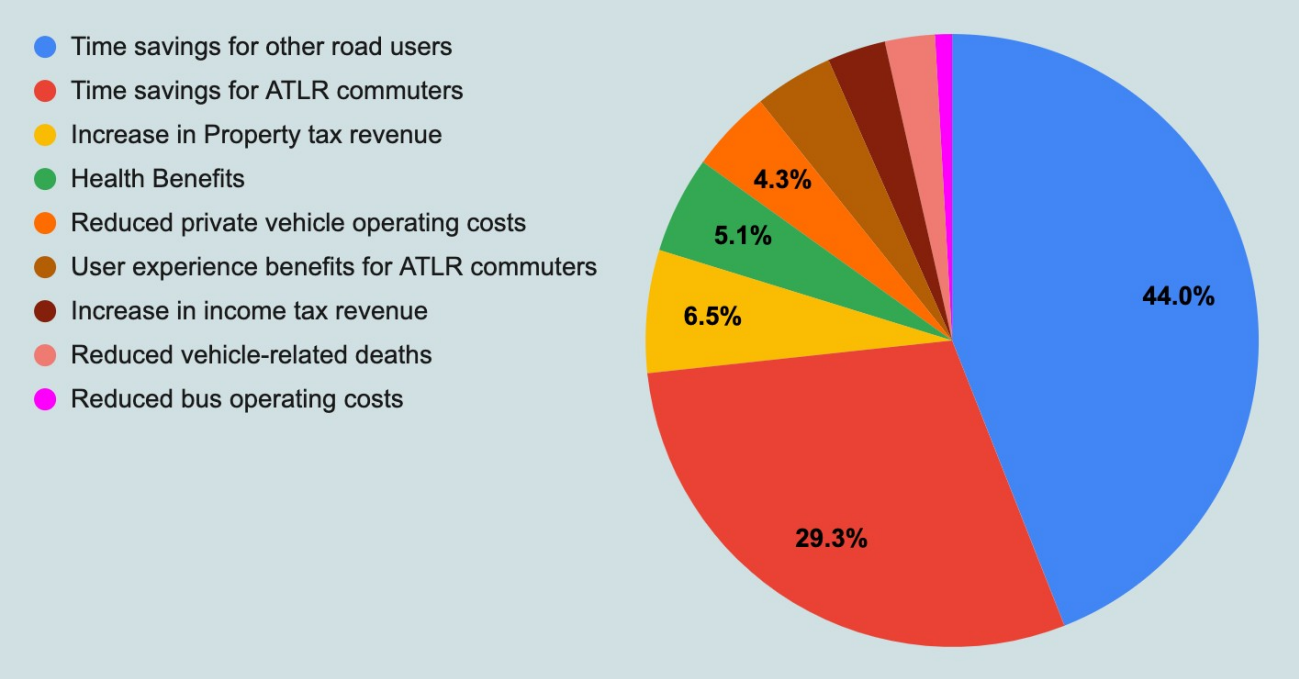
Figure 9. Breakdown of Auckland's population by local ward area

Local Board	Include/ Exclude	Population	Percent of Auckland
Rodney Local Board Area	Exclude	66,417	4%
Hibiscus and Bays Local Board Area	Exclude	104,010	7%
Upper Harbour Local Board Area	Exclude	62,841	4%
Kaipatiki Local Board Area	Exclude	88,269	6%
Devonport-Takapuna Local Board Ai	Exclude	57,975	4%
Henderson-Massey Local Board Are	Exclude	118,422	8%
Waitakere Ranges Local Board Area	Exclude	52,095	3%
Great Barrier Local Board Area	Exclude	936	0%
Waiheke Local Board Area	Exclude	9,063	1%
Waitemata Local Board Area	Include	82,866	5%
Whau Local Board Area	Include	79,356	5%
Albert-Eden Local Board Area	Include	98,622	6%
Puketapapa Local Board Area	Include	57,555	4%
Orakei Local Board Area	Include	84,318	5%
Maungakiekie-Tamaki Local Board A	Include	76,284	5%
Howick Local Board Area	Include	140,970	9%
Mangere-Otahuhu Local Board Area	Include	78,450	5%
Otara-Papatoetoe Local Board Area	Include	85,122	5%
Manurewa Local Board Area	Exclude	95,670	6%
Papakura Local Board Area	Exclude	57,636	4%
Franklin Local Board Area	Exclude	74,838	5%
Area Outside Territorial Authority	Exclude	0	0%
		1,571,715	
% of Auckland's population in Central and South Auckland			50%

Reference: Auckland Council (2019), Authors' own analysis

Annex H: Breakdown of benefits and costs of CBA

Benefits



Costs

